Absolute error

1

a 0.5 kg **b** 0.5 g **c** $5

2 $\frac{1}{2}$ mL

3

a 0.05 mL **b** 0.5 mm **c** 0.05 g **d** 0.5 s

e 0.005 kg **f** 0.05 m **g** 0.005 °C

4 We do not know what place value the number has been rounded to.

The limits of accuracy

5

a 1 cm **b** 90 cm

c 0.5 cm **d**

i No **ii** No **iii** No **iv** Yes

6

a 10 kg **b** 5 kg

c Lower bound: 65 kg, upper bound: 75 kg

7

a 50 km/h **b** 5 km/h **c** 55 km/h

8

a

i 13.455 km **ii** 13.445 km

b

i 6.45 km **ii** 6.35 km

c

i 5.5 m **ii** 4.5 m

d

i 14.005 kg **ii** 13.995 kg

e

i 4884.5 m **ii** 4883.5 m

f

i $52.50 **ii** $47.50

9 19.995 m

10 2.05 m

© ClipSAT: By Mr. Mohamed Abdallah +977597688647

11 **a** Maximum: 175, minimum: 165 **b** Maximum: 3050, minimum: 2950

12
a 26.25 **b** 0.6

13
a 13.85 m/s **b** 13.95 m/s

Percentage error

14
a ±0.91% **b** ±0.96% **c** ±7.14% **d** ±1.22%
e ±0.29%

15
a
i 4.7 kg **ii** 15.5%
b
i 0.07 m **ii** 5.3%
c
i 7.2 minutes **ii** 13.8%

16
a 15 891 **b** 16 109 **c** 18.9% **d** 19.2%
e Italy's

Area and volume calculations

17
a 153.9 cm^2 **b** 132.7 cm^2, 176.7 cm^2
c 22.8 cm^2 **d** 14.8%

18
a Upper bound: 272.6925 m^2, lower bound: 269.3925 m^2
b Upper bound: 66.2 m, lower bound: 65.8 m

© ClipSAT: By Mr. Mohamed Abdallah +977597688647

19

a Lower bound: 9.5 cm, upper bound: 10.5 cm

b 100 cm^2

c Lower bound: 90.25 cm^2, upper bound: 110.25 cm^2

d 1000 cm^3

e Lower bound: 857.375 cm^3, upper bound: 1157.625 cm^3

© ClipSAT: By Mr. Mohamed Abdallah +977597688647